do a informational math on: "quantaplasms cause photons to decay into a ground state when they are certain these quantaplasms have multiple forms of certainty and decay as one object regardless of how large they are."

The sentence you provided is not standard physics terminology and appears to be a speculative / neologistic model. I'll interpret it "informationally" in a math-plus-physics style: that is, show what structure such a statement would imply if you were to turn it into a minimal formal model, while keeping close to known quantum-mechanical language.

What the sentence seems to claim

- There is some entity called a "**quantaplasm**" that interacts with photons.
- These quantaplasms somehow **cause photons to decay into a "ground state"** under certain conditions.
- Quantaplasms have **multiple "forms of certainty"**, and they **decay as one object irrespective of size**.

In standard physics, photons in vacuum are stable and do not decay because they are massless, and their decay would violate energy–momentum and gauge-conservation constraints. Also, "ground-state photons" typically refer to virtual photons in the ground state of a light–matter system, not free photons that have "decayed" into it.[1] [2] [3] [4] [5]

Below I recast your statement into a toy model and then give an informational "math" sketch.

1. Interpret "quantaplasm" as an effective object

Suppose each quantaplasm is represented by a **quantum object** with a collective Hilbert space $\mathcal{H}_Q$, which may be multi-component (multiple "forms of certainty"). You might model this as:

$$\psi_Q \in \bigotimes_{i=1}^{N} \mathcal{H}_i$$

where each $\mathcal{H}_i$ corresponds to one "form of certainty." The overall state can be highly correlated (e.g., a many-body quantum state), so the system behaves as a **single object even when spatially large**, with global constraints:

$$\sum_i \hat{O}_i \psi_Q = \lambda \psi_Q$$

for some collective observable $\hat{O} = \sum_i \hat{O}_i$. This captures the idea that "n-body" quantaplasms still decay as one object.

2. Photon "decay into a ground state"

Assume the system is a **light–matter coupled system** (like a cavity QED or plasmonic setup), described by a total Hamiltonian:

$$\hat{H} = \hat{H}_{\text{photons}} + \hat{H}_{\text{matter}} + \hat{H}_{\text{int}}.$$

Here:

- $\hat{H}_{\text{photons}} = \sum_{\mathbf{k}} \hbar\omega_{\mathbf{k}} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}}$,
- $\hat{H}_{\text{matter}}$ describes the quantaplasms (or host matter),
- $\hat{H}_{\text{int}}$ couples them, e.g., $\hat{H}_{\text{int}} \propto \sum_{\mathbf{k}} g_{\mathbf{k}} (\hat{a}_{\mathbf{k}}^\dagger \hat{Q} + \text{h.c.})$.

In such models, the **true ground state** is not empty of photons; it can contain a finite number of *virtual* photons. A "photon decaying into the ground state" can be interpreted as:[3] [4]

1. A photon in a **bare Fock state** (excited photon mode) is absorbed or redistributed by the quantaplasm.
2. The system evolves from an excited state to , so the **photon number expectation drops**:

where $\hat{N}_\gamma = \sum_{\mathbf{k}} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}}$.

This is not literal photon decay in vacuum, but **relaxation into the ground state** of a coupled system, much like atom–cavity relaxation or emitter–plasmon relaxation.[6] [3]

3. "Multiple forms of certainty" as constraints or observables

"Certainty" can be mapped to a **constraint on probability**. For example, if an agent/quantaplasm is "certain" about some observable $\hat{O}$, then the variance vanishes:

If there are multiple "forms of certainty," you might have several operators $\{\hat{O}_1, \hat{O}_2, \ldots, \hat{O}_k\}$ such that:

$$\Delta \hat{O}_i^2 = 0$$

for each, implying the system is in a **common eigenstate** of these observables. This is typical in MBQC or many-body quantum systems where multiple conserved quantities (e.g., total energy, charge, etc.) are fixed.

Your phrase

> "they decay as one object regardless of how large they are"

then amounts to a **collective decay** or **collective relaxation** of the whole system, that can be modeled by a **super-operator**:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[\hat{H}, \rho] + \mathcal{L}[\rho].$$

The Lindblad dissipator $\mathcal{L}$ might be collective, i.e., jumps tied to a **global operator** $\hat{J} = \sum_i \hat{j}_i$:

$$\mathcal{L}[\rho] = \hat{J}\rho\hat{J}^\dagger - \frac{1}{2}\left\{\hat{J}^\dagger\hat{J}, \rho\right\}.$$

This corresponds to the whole quantaplasm decaying as one object, even if its spatial support is large.

4. Putting it into a minimalist "informational math" box

Here's a compact mathematical sketch that captures your sentence:

- Environment: a light–matter coupled system with Hamiltonian $\hat{H}$.
- Quantaplasms: quantum objects with multi-component Hilbert space $\bigotimes_i \mathcal{H}_i$, constrained to be in a **joint eigenstate** of certainties $\hat{O}_i$.
- Photons: initially in an excited state with .
- Dynamics: Markovian evolution with a **collective jump** $\hat{J}$, so the system relaxes to and the photon number expectation decays:

This is effectively your statement re-phrased as:

> **Quantaplasms (with multiple certainty constraints) act via collective dissipation that drives photons into the ground-state virtual-photon configuration.**

If you want, I can next turn this into a **toy–model Lagrangian or Lindbladian** (e.g., a small matrix example for a cavity + quantaplasm) you can plug into a numerics or QED simulation.

Would you like that done as a **minimal numerical example** (e.g., 2×2, 3×3, or 4×4 system), or as an analytic formula in a continuum QED-like field picture?

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1. https://www.reddit.com/r/AskPhysics/comments/gpkdj6/do_photons_decay/
2. https://www.reddit.com/r/askscience/comments/475evx/can_photons_decay_over_time/
3. https://pmc.ncbi.nlm.nih.gov/articles/PMC5684391/
4. https://arxiv.org/abs/math-ph/0211009
5. https://www.physicsforums.com/threads/is-photon-decay-allowed-in-qed-investigating-conservation-laws-and-amplitudes.517196/

6. https://pubs.rsc.org/en/content/articlehtml/2015/fd/c4fd00165f
7. https://daarb.narod.ru/tcp/tcpr-eng.pdf
8. https://arxiv.org/pdf/1808.03798.pdf
9. https://www.tiktok.com/@veritasium/video/7331111750420434219
10. https://www.fnal.gov/pub/science/inquiring/questions/light_dual.html
11. https://en.wikipedia.org/wiki/Ground_state
12. https://arxiv.org/html/2312.14112v1
13. https://arxiv.org/html/2410.05917v1
14. https://hal.science/hal-03477648v1/file/ROMP3---Ground-state-photon-number-at-large-distance---copie14.pdf
15. https://profmattstrassler.com/articles-and-posts/particle-physics-basics/why-do-particles-decay/most-particles-decay-why/